Assignment 1 - Research Portfolio

Draft Literature Review Chapter

Motor Skill Development In Children

Motor skills are crucial and also a fundamental part towards the overall development of children which will in turn affect their future lives. Encompassing both fine and gross motor skills that are vital for daily activities and physical health are of high importance. Research conducted by that of Gallahue and Ozmun (2006) suggests and highlights the stages of motor development and emphasises the critical periods during childhood when these skills are most effectively developed and mainly affects and develops habits that take shape into their later life. This also aligns with the notion that early childhood is a sensitive period for acquiring foundational motor competencies, which can influence later physical and cognitive development (Cools et al., 2009).

Furthermore, as suggested by Valentini and Rudisill (2004), illustrates that motor skills are not only essential for physical health but also for cognitive and social development. Their research into this topic shows that children with better motor skills are more likely to engage in physical activities, which of course contributes to better health outcomes and social integration. These are of course important and do contribute to their abilities to complete and set out their own targets. Conversely, children with poor motor skills may also experience challenges such as lower self-esteem and may even experience social isolation, which can have a long-term impact on their mental health and well being (Wrotniak et al., 2006).

Further studies conducted by Barnett et al (2016) underlines the importance of early intervention in improving motor skills. Further evidence from Cariney et al. (2010) also supports this and they argue that structured programmed developed for focusing on motor skill development can indeed lead to significant improvements in children's physical competence and confidence, which are critical for lifelong physical activity participation. The importance of this is often underestimated by many people and overlooked in children by adults. Therefore, this indeed underscores the need for implementing effective tools and possible programs that can in fact engage children in their motor skill development. This is of particular importance in an era such as today's where a sedentary lifestyle is increasingly becoming the norm and becoming ever more common (Tremblay et al., 2011).

Including Gamification In Education

The thought and concept of including gamification in education involves the likes of incorporating game design elements into non-game context. This is done with the

aim of enhancing engagement and motivation as researched by Deterding et al., 2011. In typical educational settings, including gamification has been proven and shown to increase student motivation, participation and arguably most importantly retention of information as suggested by (Hamari, Koivisto, & Sarsa, 2014). This is vital in terms of goal setting and motivation towards achieving the goals. Hand in hand with this is the Self-Determination-Theory (SDT) as devised by Deci and Ryan, 2000 also provides a framework for understanding how gamification can essentially satisfy basic psychological needs which are the likes of competence, autonomy and relatedness. All of these are useful and essential for intrinsic motivation. Intrinsic motivation is essentially the desire of an individual to engage in a particular behaviour because of the inherent satisfaction of that particular activity rather than their own personal desire for a reward or a specific outcome (Cherry, 2023).

Research conducted by Dominguez et al. 2013 demonstrated that gamified learning environments could in fact lead to higher engagement levels and thus lead to better learning outcomes.. Their study involved a gamified course in higher education, which found that students who participated in this gamified version were actually more engaged and performed better in assessments compared to those in a traditional learning environment. This then suggests that gamification can make learning more enjoyable and effective by leveraging elements such as ‘like points’, badges, leaderboards and even potentially comparisons with friends or peers. Which could in fact bring a competitive edge to further boost engagement levels.

Brom, Preuss and Klement (2011), explored the impact levels of gamification on learning in younger students. They found that game-based learning could improve attention, memory and cognitive flexibility. This is also due to the fact that gamification leads to higher engagement due to the simple fact it could be seen as more interesting and potentially rewarding too. The use of games and gamification in education however is not without challenges. Issues such as the likes of balancing educational content with entertainment and ensuring that these game element designs do align with the intended learning objectives and critical considerations (de Freitas, 2006). These studies notably highlight the potential of gamification to transform education experiences, thus again making them more interactive and engaging which is also particularly relevant for developing motor skills in children as talked about earlier.

Technology - Assisted Learning Tools

Technology is ever growing and increasing and becoming further enhanced at a rapid rate and has become an integral part of modern education and life as a whole. This is due to the fact they offer many innovative ways to support learning and development. Technology-assisted learning tools range from the likes of interactive software to mobile applications and each are designed to facilitate and offer different views and aspects of learning. The integration of technology in education has been

shown and proven to provide personalised learning experiences, which on its own can enhance individuals' learning capabilities and also accommodate different learning styles and also offer immediate feedback (Livingstone & Helsper, 2007)

A study done by Papastergiou (2009) focused on and examined the effectiveness of digital games within education and found out that they could significantly enhance students' learning experiences by providing interactive and engaging content. The study focused on a computer game designed for the purpose of teaching computer science concepts and revealed that students who used the game as a method of learning did in fact show improved understanding and retention compared to those individuals who received the traditional instruction and method of learning. This evidence thus supports the idea that digital games and apps within an educational setting can cater to individuals' learning styles and can also serve as a powerful tool for teaching complex concepts especially in a more accessible and engaging manner compared to the more traditional methods.

Building upon this, the use of mobile apps within an educational setting, particularly in developing motor skills - is in fact gaining traction. Apps such as the likes of 'ABCmouse' and 'GoNoodle' as well as even more traditional websites such as 'BBC Bitesize' have been used extensively to support early childhood and older childs' education and physical activity, respectively. These sorts of tools leverage interactive content, gamification and also multimedia amongst others to enhance learning outcomes and build upon the traditional learning methods. A review focused on by Sutapa et al., (2021) found that such apps could significantly improve a child's motor skills, cognitive abilities and their engagement levels further.

However, opposite to this there are challenges associated with the use of technology in education too, including issues related to access, equity and also the quality of content available. It is argued by critics that whilst technology can offer innovative solutions it must also be carefully monitored, designed and implemented to avoid widening the digital divide further and also ensure it meets educational standards too (Selwyn, 2011). These considerations are crucial when focusing on the design of the Targetomoter App, as they emphasise the need for a focused, well-structured, inclusive and accessible tool above all else.

The Targetomoter App: Addressing Potential Gaps and Opportunities

The Targetomoter App is going to be designed to address the specific needs for engaging, effective and accessible tools for developing motor skills in children and of course if it is capable of that it'll likely be effective for adults too. The current literature on motor skill development and gamification highlights the potential benefits of integrating these approaches, but also points to the gaps in the existing solutions. For instance, whilst many apps and programs do focus on cognitive development or

general physical activity, there is a noticeably large lack of specialised tools that target finer and gross motor skills in a structured, logical and gamified manner.

The existing research emphasises the importance of having tailored interventions that can adapt to an individuals' learning needs and preferences suggested by (Shute & Ke, 2012). The Targetomoter app aims to fill this gap by providing a customisable experience that adjusts to the child's progress, thus offering challenges that are appropriate for their skill level. This sort of personalised approach may not only enhance engagement but can also ensure that the learning experiences offered are actually effective and enjoyable too.

Additionally, the Targetomoter apps use of any real-time feedback and tracking aligns with best practices in educational technology, thus offering valuable insight into the child's progress and areas for improvement (Hattie & Timperley, 2007). This sort of feature is particularly important for parents and also educators who play a critical role in supporting the child's development. By providing detailed reports and analytics, the app helps stakeholders make informed decisions about the child's learning journey. The Targetomoter app also aids in addressing concerns by being available on multiple platforms including the likes of smartphones and tablets, thus making it accessible to a broader audience. This sort of inclusivity is crucial in ensuring that children from diverse backgrounds can benefit from this app, regardless of socioeconomic status or geographical location (UNESCO, 2015) The Targeomoter app's design also considers user-friendliness, along with intuitive navigation and engaging visuals that appeal to young children.

Motor Skill Development In Adults

In adults motor skills encompass a range of abilities that range from skills such as walking and lifting to fine motor skills like typing and precise hand movements. These sorts of skills are vital for maintaining independence and performing occupational tasks efficiently. As adults age, there is of course a natural decline in these abilities which can be further exacerbated due to factors such as sedentary lifestyles, injuries and chronic conditions. Studies have shown that target exercise can mitigate these declines and improve quality of life and prolong independence.

The likes of wearable devices are being increasingly used to track and promote motor skill development. These technologies often incorporate sensors and biofeedback mechanisms which thus provide real-time data and personalised feedback to users. Wearable devices can also effectively monitor and enhance the fine motor skills of individuals with hand tremors and offer non-invasive solutions for managing symptoms.

Theories Of Goal Setting And Motivation

In recent years, goal-setting and motivation have emerged as critical factors in personal and professional development as well. The Targetomoter app aims to support individuals in setting and achieving their goals by leveraging technology to provide structured guidance, motivational reminders and also potentially gamified incentives. Expanding on previously discussed literature reviews, further examined will be that of theoretical and empirical research on goal setting, motivation, digital interventions and gamification. This also provides a foundation for this app's design and highlights the importance of these elements mentioned in enhancing user engagement and effectiveness in goal attainment.

Goal-setting theory which was pioneered by Locke and Latham (1990), suggests that specific, challenging goals can significantly enhance performance. The theory provides insights into the suggestion that clear goals can help focus attention, mobilise effort and also increase further persistence. In a meta-analysis conducted by Locke and Latham (2002) it was found that goal-setting had a robust effect on performance across various tasks and settings, thus underscoring its universal applicability.
In relation to digital interventions, goals can be used for tailoring user experiences and also enhancing engagement and outcomes. For example, apps designed for fitness or productivity often include goal-setting features to encourage users to set specific targets, track progress and achieve their desired outcomes (Deci & Ryan, 2000). These sort of applications benefit from incorporating SMART goals which stand for specific, measurable, achievable, relevant and time bound. This is essentially used to help increase the likelihood of success.

Further building on digital interventions, such as apps and online platforms and even websites, these have become increasingly popular for facilitating goal-setting and achievement. Research undergone by Michie et al (2011) has shown that digital tools can effectively support behaviour change through mechanisms such as self-monitoring, feedback as well as social support. The Targetomoter app can aid in leveraging these mechanisms to help users set and achieve personal and professional goals. Apps that include the likes of goal-setting features often tend to utilise data analytics to provide personalised feedback and insights. For instance, productivity apps might analyse a particular user’s work patterns to suggest optimal time for focus and rest, whilst fitness apps could adjust workout plans based on performance metrics instead as posed by (Fogg, 2009). These sorts of personalised elements are genuinely crucial in maintaining high user engagement and their own personal motivation.

In the context of motivation and motivational reminders. These could be an essential and important component of The Targetomoter app and would be designed to sustain user engagement and commitment to their goals. According to research by Kessler et al (2016) suggests that periodic reminders can significantly increase an individuals’ adherence to their goals, particularly when these reminders are positively

framed and provide actionable advice to boost motivation and engagement. Furthermore, the use of notifications and reminders particularly in digital interventions is well-documented, with many various studies providing ample evidence showing that timely prompts can in-fact boost engagement and prevent drop-offs in user activity as well (Eysenbach, 2005). The Targetomoter app can also employ this sort of strategy by sending regular, motivational messages to users, thus encouraging them to stay focused on their goals.

Relating back to that of gamification and the rewards system touched on earlier and incorporating this into the Targetomoter app can further enhance user motivation and engagement. Gamification involves integrating game-like elements, such as points, badges and leaderboards into this non-game context in order to make ordinary tasks a lot more engaging. Research suggests that gamification can again significantly increase motivation and participation in a variety of activities from education, to health and fitness too. A well thought out and well designed rewards system can also provide extrinsic motivation. This is simply motivation that is derived from the wanting of receiving a reward for a particular behaviour. Encouraging users to persist in their efforts. For instance completing a particular goal could unlock new app features or reward users with spendable ‘points’ in the app for different features, earn discounts or provide virtual rewards per say, which can indeed be extremely powerful motivators. The Targetomoter app could utilise these strategies to maintain long-term user engagement and also promote consistent goal achievement.

The integration of technology in personal development has also opened several various new avenues for supporting individuals in their pursuit of their goals. The likes of wearable technology, mobile apps and online platforms also offer unprecedented access to personalised feedback and real-time tracking of their progress (Pew Research Center, 2015). The Targetomoter app, by incorporating these sorts of features and adding and using advanced data analytics and user personalisation can produce tailored advice and feedback, thus enhancing users’ abilities to achieve their goals once again. Moreover, the use of artificial intelligence (AI) and machine learning (ML) can offer adaptive learning techniques and experiences, where the app adjusts its recommendations based on the user’s progress and preferences. This approach not only incorporates and increases the relevance of the feedback provided but also ensures that the user’s experience is continuously optimised for better outcomes (Bickmore et al., 2005).

This literature review essentially highlights the critical need for innovative solutions in motor skill development and particularly tools that are capable of leveraging gamification and technology. The Targetomoter app also stands out as a promising solution in addressing the gaps identified in existing research and also offers a comprehensive approach to enhancing children's motor skills. Additionally, by drawing on the strengths of gamification, technology-assisted learning and also personalised education, the app has the potential to make a significant impact on not

only children's physical health and capabilities but also this cognitive and social developments. The inclusion of multiple literature sources again not only strengthens the argument for the Targetomoter apps development but also provides a well rounded structure allowing for the understanding of the current landscape. This comprehensive review thus sets the stage for the development and also implementation of the Targetomoter app, providing a solid foundation for further research and practical applications too.

In the context of goal setting and motivation the literature review also highlights the importance of doing so, for motivation and using the likes of digital interventions and gamification in order to enhance personal development and enable successful goal achievement. The Targetomoter app with its focus on goal setting and motivational reminders as well as potential gamified elements is in a well positioned area to leverage these sorts of insights. By also integrating proven strategies supported by ample evidence from a large variety of sources and literatures, the Targetomoter app can provide a comprehensive tool for users that are looking to achieve both their personal and professional goals. This literature review chapter thus provides a solid foundation for the apps development and ensures it is grounded around robust theoretical and empirical research.

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Personal Reflection on Early Stages of the Project Process

Initial Vision and Evolution

The initial vision for the Targetomoter app was centred around enhancing motor skills in children using gamification techniques. However, as I delved deeper into the research and planning stages, it became clear that the app's potential could be

broadened to encompass a wider audience. The shift from a focus on children to a more inclusive target demographic of adults seeking personal development and goal achievement arose from a reevaluation of the app's core objectives and the potential for a broader impact.

This transition required a significant change in the research questions and objectives. Initially, the research question revolved around the efficacy of gamification in improving motor skills among children. However, the refined question now focuses on how digital tools can facilitate goal setting and motivation across different life stages, particularly for adults. This shift not only aligns better with a broader market need but also taps into a more diverse user base.

Research Methodology and Adaptations

The research methodology also evolved alongside these changes. Initially, the project aimed to use experimental designs to measure improvements in motor skills among children using gamified exercises. The new approach, however, incorporates a mixed-methods design, combining quantitative surveys and qualitative interviews to gather data on user engagement, motivation levels, and the perceived effectiveness of the app's features, such as goal-setting frameworks and motivational reminders.

This change was driven by the realisation that understanding the nuances of user motivation and the psychological factors influencing goal achievement required more than just quantitative measures. The qualitative component will provide deeper insights into user experiences and the contextual factors that affect their engagement with the app. This methodological shift ensures a more holistic evaluation of the app's impact.

Adjustments in Evaluation Approaches

The evaluation approaches also saw significant refinement. Initially, the evaluation was primarily focused on assessing motor skill improvements through pre- and post-intervention assessments. However, with the shift in focus, the evaluation now emphasises a broader range of outcomes, including psychological metrics such as self-efficacy, motivation, and user satisfaction. The evaluation will also consider user retention rates and the app's effectiveness in helping users achieve their goals over time.

To support these new evaluation methods, the app will now include features like in-app surveys and progress tracking tools, allowing users to provide ongoing feedback and self-assessments. This continuous data collection approach will enable a more dynamic and real-time evaluation of the app's effectiveness, offering valuable insights that can be used for iterative improvements.

Reflections and Future Directions

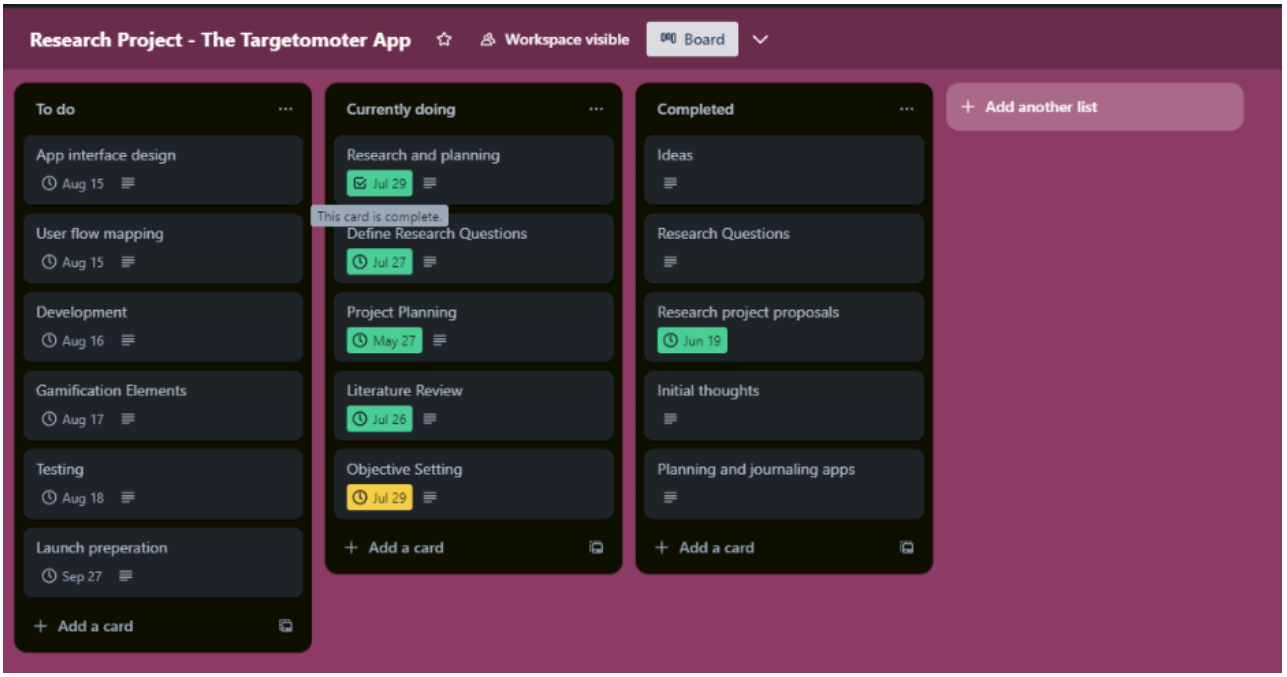
Reflecting on the early stages of the project, the most significant learning has been the importance of flexibility and adaptability in research. The shift from a child-focused motor skill development app to a more inclusive goal-setting and motivation tool was not just a change in audience but also a fundamental realignment of the project's purpose and scope. This experience highlighted the importance of aligning research objectives with real-world needs and market potential.

Furthermore, this shift underscored the value of a comprehensive literature review and stakeholder feedback in shaping the project's direction. The expanded literature review not only informed the app's features and design but also provided a solid theoretical foundation for understanding the psychological principles underlying goal-setting and motivation.

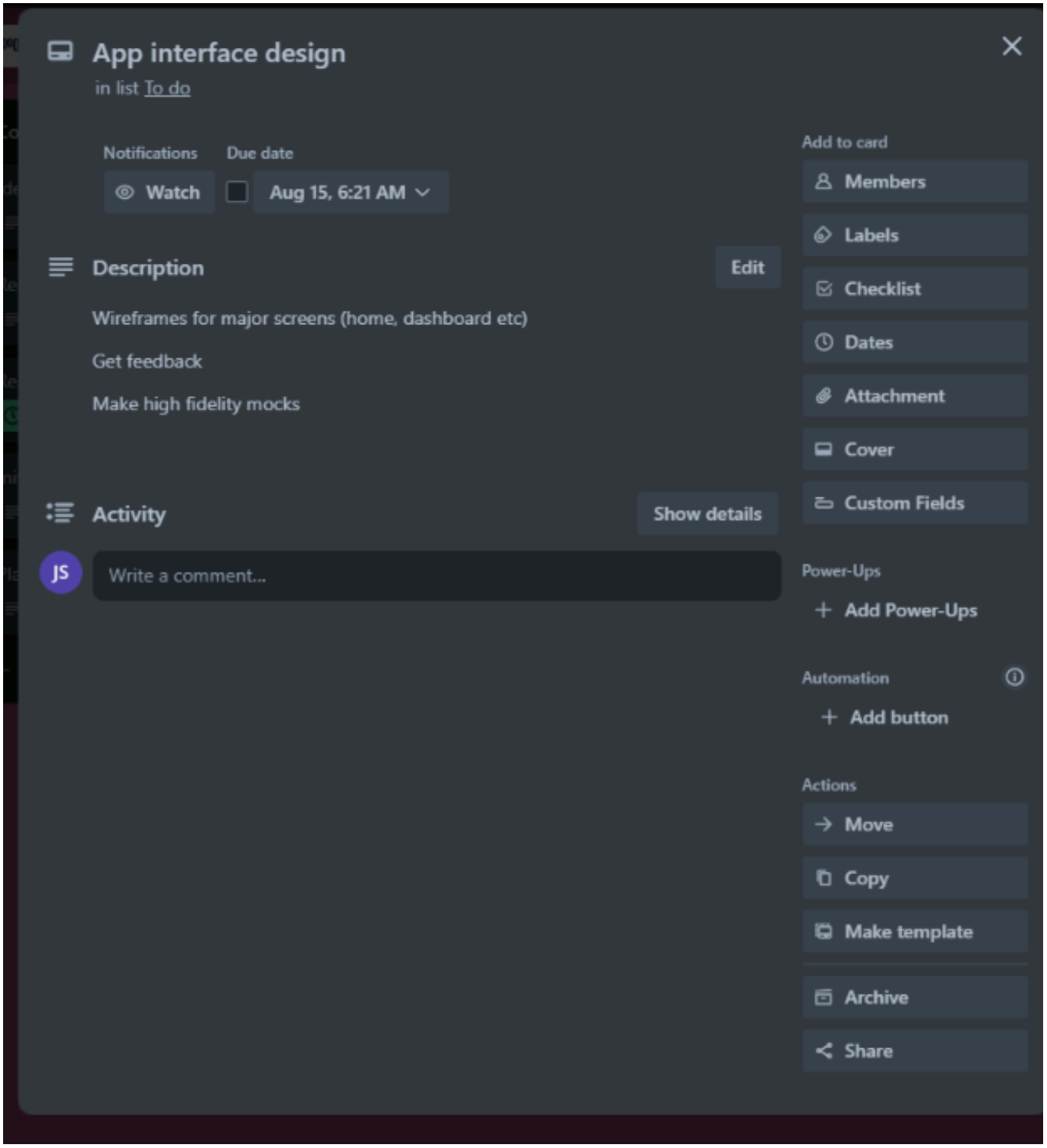
Moving forward, the focus will be on refining the app's features to cater to the diverse needs of adult users, including considerations for accessibility and inclusivity. The project will also explore integrating emerging technologies like artificial intelligence for personalised goal recommendations and reminders, enhancing the app's ability to support users in their personal development journeys.

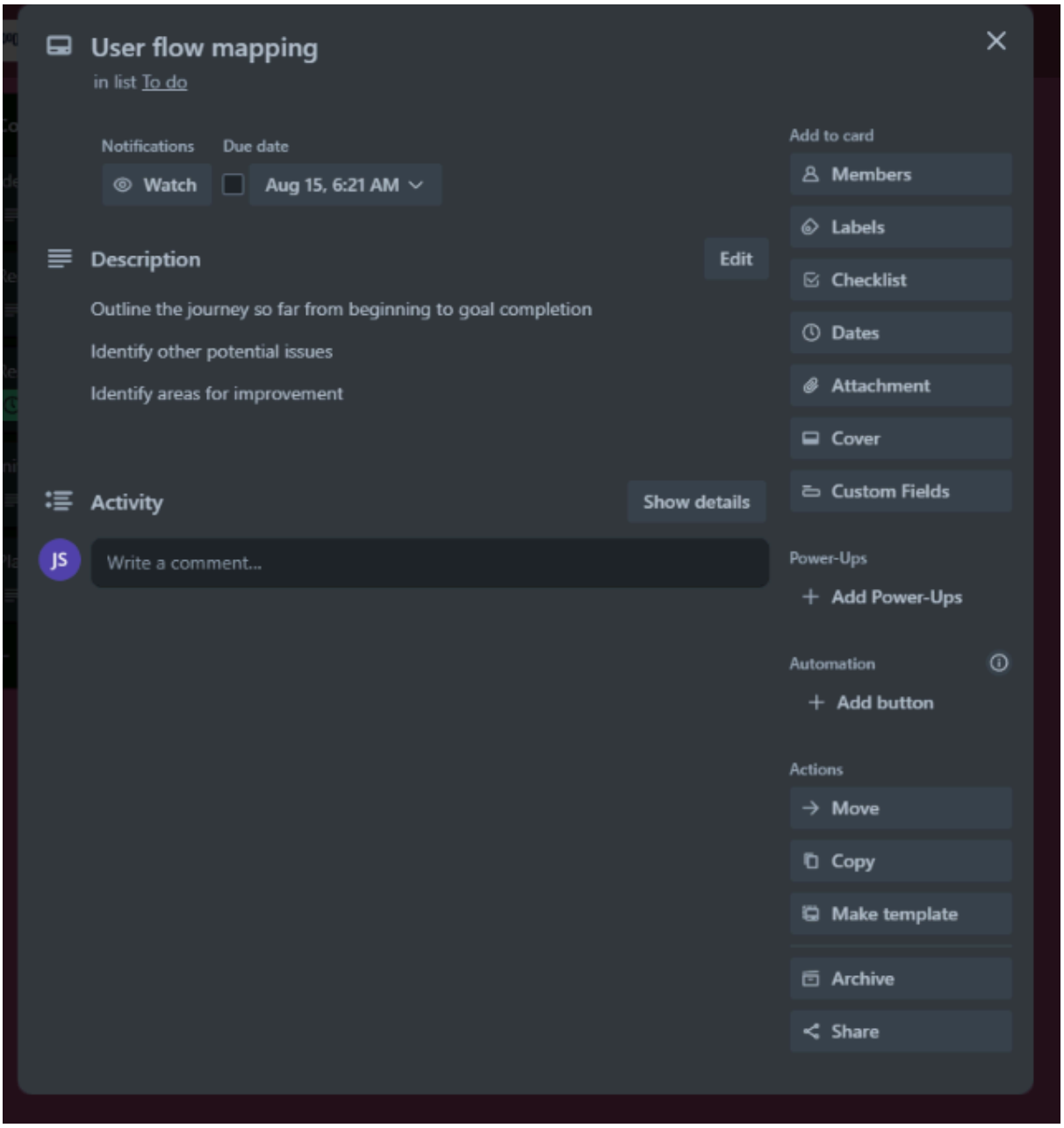
In conclusion, the early stages of the Targetomoter project have been a period of significant learning and adaptation, setting a robust foundation for the app's development and ensuring its relevance and impact in a broader context.

<u>Research progress to date</u>



I have been using Trello to keep a track of certain tasks which require completion and my attention and to keep a note of what I have already done and what needs to be done. As seen I have set timelines of when I'd like tasks to be completed and whether they have been completed or not. Keeping the overview clean and simple allows me to stay on top of the workloads and easily keep track of where I am up to.





Within each card along with the due dates I will add additional information just to aid with any brainstorming or these are just initial ideas and thoughts related to that particular section. This is useful as it allows me to remember past thoughts or aid myself in remembering what must be done for each section.

Below is the initial timeline thought up during the implementation and delivery during the previous module assessment. The project as mentioned below was estimated to take around 1-2 months in total.

It is probably thought to potentially take between 1-3 months now, however the initial estimate wasn't too far off.

Project Schedule

The project schedule is a general outline to the key tasks, timelines and possible deliverables required in order to develop and evaluate the Targetomoter app. The project is planned over a period of between 1-2 months. The project schedule is split up into distinct phases which are the following; Planning, Development, Testing, Evaluation and Final Reporting.

Detailed project schedule timeline:

- **Planning Phase**

- Task: Define the project scope and objectives of the project
- Conduct a literature review
- Focus on designs and initial wireframes
- Design potential prototypes
- Deliverable = Project Plan outline document

- **Development Phase**

- Task: Develop core features of the project
- Task: Conduct and integrate any user feedback from initial prototypes
- Task: Implement User interface or user experience design
- Deliverable = Potential Alpha version of the App

- **Testing Phase**

- Task: Conducting usability testing
- Task: Identifying and fixing any issues or bugs
- Task: Refine the app further based on test results and any form of data received
- Deliverable = Beta version of the Targetomoter App

- **Evaluation Phase**

- Task: Deploy app to test group
- Task: Collect and analyse user data
- Task: Conduct qualitative interviews
- Deliverable = Evaluation Report

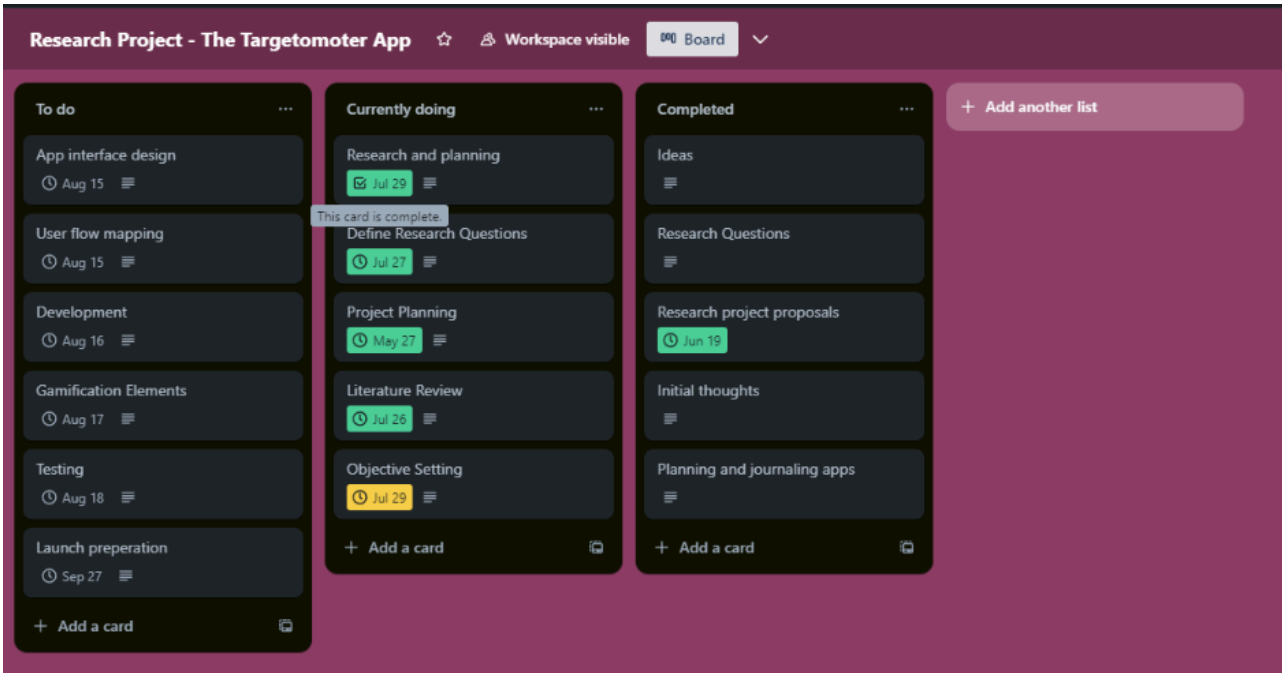
- **Final Reporting Phase**

- Task: Compile final report
- Task: Present all findings
- Deliverable = Final Project Report and presentation

- **Milestones and Deliverables**

- Project plan document
- Alpha Version of the Targetomoter App

Supporting Evidence (Appendices)



This evidence from Trello just showcases what I have done as well as what needs to be done and the progress of tasks currently being done. This just documents the project's scope, objectives and key elements as well as goals and major deliverables.

This also documents the task lists with the different lists I have used. It also documents the completed elements amongst others.

Full Reference List for literature review summary

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Consent Form

Title: Enhancing Goal Achievement Through The Targetomoter App

Investigator: Jaq Steele

Organisation: University Of Sunderland

Contact Information: Jaqsteele01@hotmail.com

Date: ……

Introduction:

You are invited to participate in a research study conducted by Jaq Steele at the University of Sunderland. The purpose of this study is to evaluate the effectiveness of the Targetomoter app in helping individuals set and achieve personal goals. This form provides the information about the study and seeks your consent to participate.

Purpose of the study:

The goal and purpose of this study is to assess how using the Targetomoter app, which includes features such as goal tracking, motivational reminders and a rewards system, impacts participants' goal-setting and achievement behaviours.

Procedures:

As a participant you will be asked to:

- Use the Targetomoter app to set and track specific goals over a given time period
- Respond to initial and follow up surveys about your experiences and progress
- Participate in a brief interview and if you agree, to further follow up insights into your experiences.

Participation involves minimal risk and your feedback will contribute hugely to the development of the app and understanding its effectiveness.

Confidentiality

All information collected within this study will be kept confidential and your data will remain anonymised.

Voluntary Participation

Your participation is voluntary and you have the right to withdraw from the study at any time without penalty. You are also entitled to not answer any questions and not participate in any parts of the study you don't wish to.

Potential Benefits

No direct benefits however, your help with aid in making the app better and also if you decide to continuously use the app, potential new features or premium features / rewards will be given free of charge.

Potential Risks

There are minimal risks associated with this study except potentially the possible discomfort of discussing personal goals. You are of course free to skip any questions you do not feel like answering.

Informed Consent

By signing below you agree to participate within this study and understand all the information provided and are at least above the age of 18. You voluntarily agree to participate.

Participants signature …….. Date ……

Participants Printed Name ……..

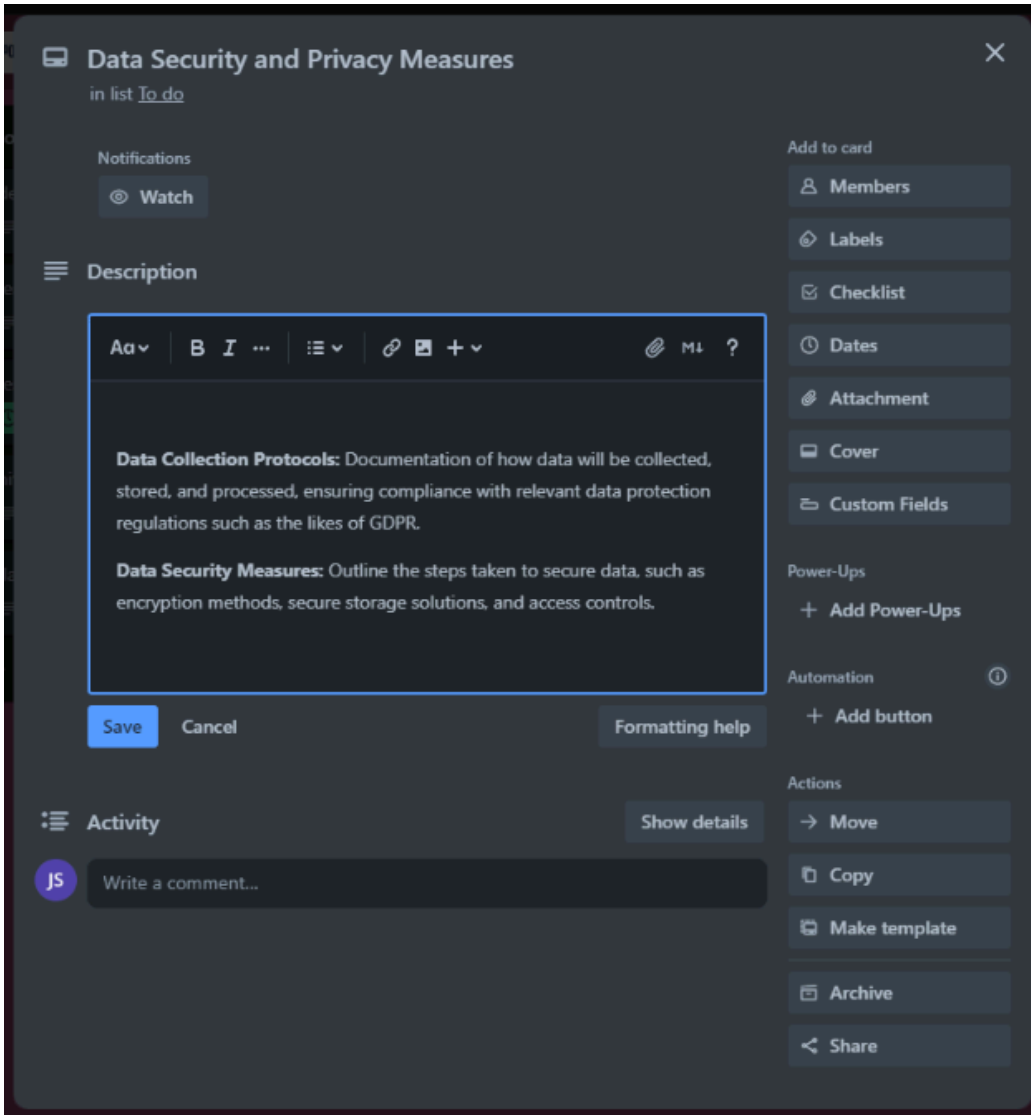
Researcher's Signature ……

Date …….

If you have any additional information or questions feel free to reach out to the contact information provided above.

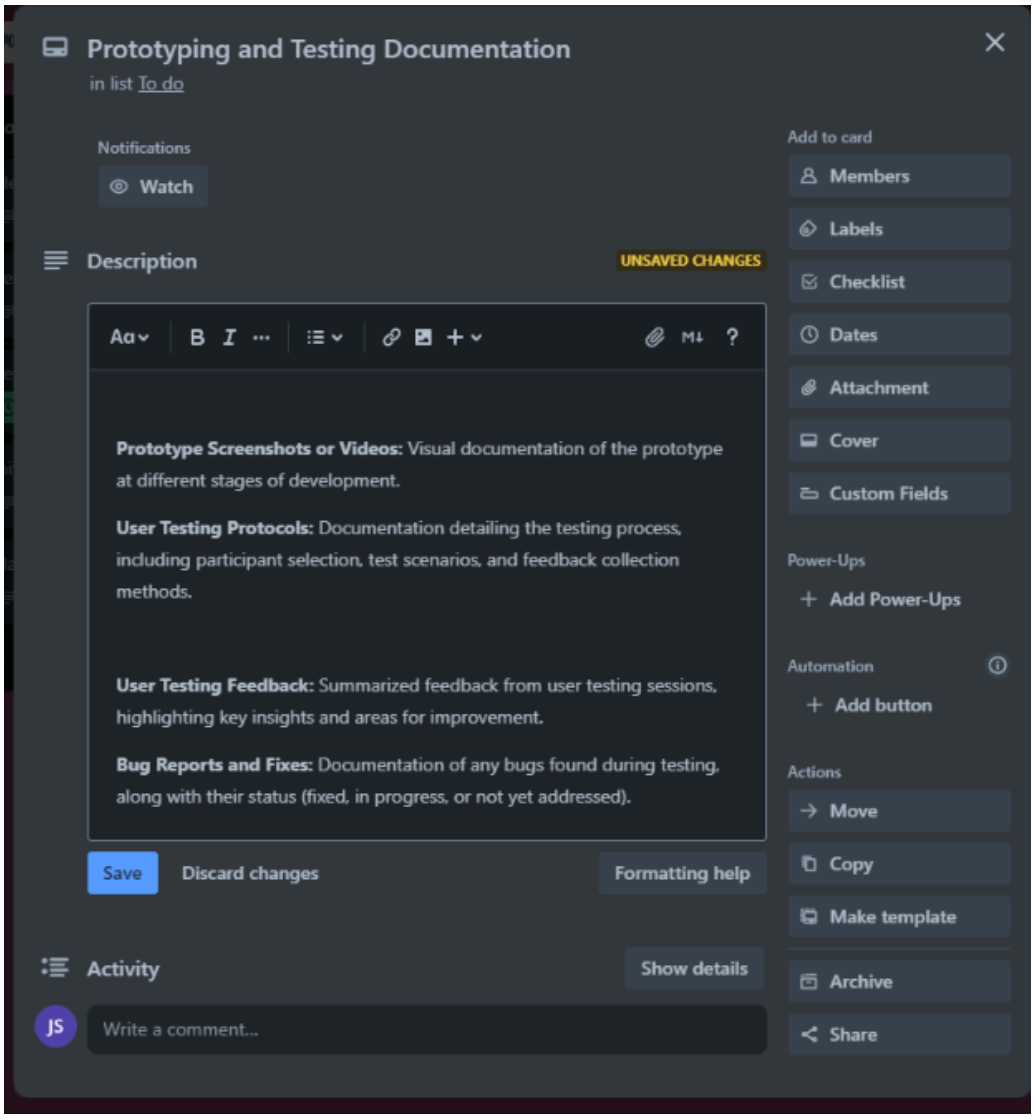
<u>Data Security and Privacy Measures</u>

This section is a focus for the future and is within the To do section. The image below details what needs to be done for this section.



Prototyping and Testing documentation

This section also indicates a future section which will be worked on and below in the image it shows the possible tasks for that section.



Surveys and Questionnaires

Information regarding user feedback and usability testing will likely be included in the surveys and questionnaires. This requires the further development of this app in order the ensure the questions align with the app at each stage as it will vary and change over time.